

Akarsh Vijjapu

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PROFESSIONAL SUMMARY

Backend-focused Software Developer pursuing B.Tech in Computer Science Engineering (CGPA: 8.86) with practical experience building scalable web applications using Python, FastAPI, PostgreSQL, and SQLAlchemy. Developed and deployed three end-to-end projects featuring REST APIs, real-time WebSockets, AI-integrated chatbot functionality, JWT authentication, and role-based access control. Solved 100+ Data Structures and Algorithms problems on LeetCode and HackerRank, demonstrating strong problem-solving skills.

TECHNICAL SKILLS

Languages: Python, Java

Database: PostgreSQL, SQLAlchemy ORM, SQL

Deployment: Render, Vercel, Git, GitHub

Backend: FastAPI, REST API Design, WebSockets, CRUD Operations, Background Tasks

Frontend: HTML5, CSS3, JavaScript

Core CS: Data Structures, Algorithms, OOP

PROJECTS

MediQueue — Hospital Queue Management System

github.com/akarsh277/MediQueue

Python · FastAPI · PostgreSQL · SQLAlchemyORM · JWTAuth · RBAC · Vercel · Render

medi-queue-project.vercel.app

- Architected a queue management backend with full CRUD operations for patient records, appointments, and token assignments using PostgreSQL and SQLAlchemy ORM.
- Implemented JWT-based authentication and role-based access control (RBAC) across Admin, Doctor, and Patient modules with strict permission enforcement per role.
- Deployed frontend on Vercel and backend on Render with PostgreSQL database, configuring CORS and environment-based API URLs for cross-origin communication.

DisasterLink — Real-Time Emergency Operations Platform

github.com/akarsh277/Disaster

Python · FastAPI · PostgreSQL · SQLAlchemy · WebSockets · JWTAuth · Vercel · Render

disaster-link-omega.vercel.app

- Engineered a real-time disaster coordination platform with persistent WebSocket connections and broadcast logic pushing live alerts to citizens, volunteers, and authorities simultaneously.
- Implemented JWT-based authentication for volunteer and authority login; applied rate limiting on report submission endpoints to prevent spam during high-traffic events.
- Migrated database from SQLite to PostgreSQL for production deployment; built event-driven backend with background tasks for alert escalation without polling overhead.

CureBot — AI-Powered Medical Assistant

github.com/akarsh277/CureBot

Python · FastAPI · LLMAPIIntegration · Vercel · Render

cure-bot-phi.vercel.app

- Built an AI symptom assessment chatbot backend integrating an LLM API for automated symptom responses and health suggestions.
- Implemented response caching to reduce redundant AI API calls and structured background tasks for async AI response logging without blocking the main request thread.
- Deployed frontend on Vercel with backend hosted on Render, connected via REST API.

EDUCATION

B.Tech — Computer Science Engineering — Pragati Engineering College, Surampalem

2024–2028 · CGPA: 8.86

Intermediate — MPC — Narayana Junior College, Kakinada

2022–2024 · 94%

SSC — Sama School, Jagannaickpur

2021–2022 · 96%

CERTIFICATIONS & ACHIEVEMENTS

- NPTEL IIT Kharagpur — Programming in Java
- Scaler Academy — Essentials in Python
- IBM Skills Network — SQL & Relational Databases
- Be10x — AI Tools & ChatGPT Workshop
- Brain O Vision — Web App Development Workshop & 24Hr Hackathon
- Participant — MindSprint 2K25 Hackathon
- Participant — Srujana Hackathon
- Solved 100+ algorithmic problems on LeetCode and HackerRank